

Social Media Mining Assignment-1

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Abstract

Toxic comment detection is essential for maintaining safe online environments. In this assignment, I developed two machine learning models—a traditional Support Vector Classifier (SVC) and a transformer-based BERT model to classify social media comments as toxic or non-toxic. I used a dataset of 4000 comments from Reddit, Twitter/X, and YouTube, applying text cleaning, tokenization, and label aggregation through majority voting. Both models were trained and evaluated using accuracy, precision, recall, and F1-score. Experimental results show that the BERT model significantly outperforms the SVC model, achieving a higher F1-score and better recall, highlighting the superiority of transformer-based approaches for text classification. In conclusion, while the BERT model demonstrated superior performance, the SVC model remains a reliable and efficient alternative for less resource-intensive deployments.

1. Introduction and Background

Social media platforms are crucial spaces for public discourse, but they often face the challenge of harmful interactions. Toxic comments, which include offensive, abusive, or disruptive language, contribute to negative experiences and can drive users away from online communities. Identifying and moderating such content is vital for maintaining healthy online environments. However, manual moderation is time-consuming and subjective, which motivates the use of approaches through machine learning.

This report evaluates how machine learning models perform in detecting toxic comments by implementing and comparing two approaches: a traditional Support Vector Classifier (SVC) and a transformer-based BERT model. Text classification for toxicity detection is a key task in natural language processing (NLP). Traditional models such as SVC rely on feature extraction techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) to numerically represent textual content.. In contrast, modern transformer models like BERT use contextual embeddings, enabling them to capture nuanced meanings within text.

The dataset used in this assignment consists of 4000 social media comments annotated by multiple human reviewers. Labels were aggregated using majority voting to ensure reliability. Both models were trained and assessed through standard metrics such as accuracy, precision, recall, and the F1-score to gauge model performance.. This assignment not only demonstrates technical implementation but also highlights the advantages and limitations observed in each model's performance on toxic comment detection. Additionally, it emphasizes the importance of balancing recall and precision to effectively detect harmful content while minimizing false positives.

2. Dataset and Methods

In this part, I will dive into the dataset and methods that helped shape the work. We gathered comments from popular social media platforms. Each comment comes with labels that indicate whether it's toxic or not, thanks to the helpful input from humans who annotated them. To get the data ready for analysis, we cleaned it up, combined the labels, pulled out important features, and then set everything in motion with our model implementation.

2.1 Dataset

In this section, I describe the dataset provided for my assignment and the methods I used to prepare it for analysis. The dataset consists of social media comments collected from platforms such as Reddit, Twitter/X, and YouTube. The dataset was separated into two segments: one for training and another for testing.

The training dataset contains 4,000 social media comments, each reviewed by five human annotators. These annotators provided labels indicating whether a comment was toxic or not. To determine the final label for each comment, a majority voting system was applied: if three or more annotators marked the comment as toxic, it was labeled as toxic; otherwise, it was labeled as non-toxic. This method ensured that the labels were consistent and reflected a consensus rather than individual opinions.

Unlike the training dataset, the test set does not include labels, allowing it to be used for model evaluation. The test dataset has the same structure as the training dataset but excludes the `composite_toxic` field, which contains human annotations.

Both datasets share a common structure and include several fields that provide context for each comment. The `text` field contains the comment itself, serving as the primary input for classification. The `parent_comment` field indicates if the comment is a reply, showing the comment it responds to or null if it is an independent comment. To provide additional context, the `article_title` field contains the title of the news article where the comment was posted, and the `article_url` field links to the news article. Additionally, the `platform` field specifies where the comment was posted (e.g., Reddit, Twitter, or YouTube), while the `platform_id` provides a unique identifier for the comment on the platform, which could be useful for external reference or further data collection.

The `composite_toxic` field, present only in the training dataset, contains five annotations from human reviewers indicating whether they considered the comment toxic (true) or non-toxic (false). The final label used for model training, stored in the `is_toxic` field, was derived from these annotations through majority voting. After generating the final labels, the `composite_toxic` field was removed from the dataset, retaining only the binary `is_toxic` label for simplicity.

2.2 Data processing

In this section, I describe the preprocessing steps I applied to the provided dataset to prepare it for model training. Proper data preprocessing is essential for ensuring that machine learning models can effectively learn patterns from the data. I focused on three main tasks: text cleaning, label aggregation, and feature extraction.

The first step was text cleaning, where I standardized the comments by converting all text to lowercase to ensure consistency. I removed punctuation, special characters, and extra spaces to reduce noise and simplify the input for the models. Additionally, I stripped any emojis and non-ASCII characters, as these elements could confuse the models without adding meaningful information.

Next, for the training dataset, I performed label aggregation to create a final binary label for each comment. The training dataset included `composite_toxic`, which contained five annotations from human reviewers indicating whether a comment was toxic or not. To consolidate these annotations, I applied a majority voting system, classifying a comment as toxic if three or more annotators labeled it as such. From this aggregation, I created a new `is_toxic` column to serve as the target label for model training. After generating the final labels, I removed the `composite_toxic` column to simplify the dataset and avoid redundancy. To simplify classification, I labeled toxic comments as '0' and non-toxic comments as '1.' This binary scheme was used consistently across both SVC and BERT models.

Finally, I performed feature extraction, which involved converting the cleaned text into numerical representations suitable for model input. I used different approaches for each model. For the SVC model, I utilized the TF-IDF (Term Frequency–Inverse Document Frequency) method, which measures the significance of words by analyzing their occurrence within individual comments relative to their rarity across the entire dataset. I limited the vocabulary size and removed stop words to reduce noise and improve model efficiency. For the BERT model, I used the BERT tokenizer, which transformed each comment into tokens, input IDs, and attention masks. I also applied padding and truncation to ensure that all tokenized inputs had the same length, which is crucial for batch processing. The attention masks indicated which tokens should be considered during training and which were padding.

Through these preprocessing steps, I ensured that the dataset was clean, structured, and optimized for training both the SVC and BERT models. These preparations were critical for enhancing the models' ability to detect patterns in the text and classify comments as toxic or non-toxic accurately.

2.3 ML methods

In this section, I describe the machine learning models I implemented to classify comments as toxic or non-toxic. I used two approaches: a traditional machine learning model, Support Vector Classifier (SVC), and a deep learning model, BERT (Bidirectional Encoder Representations

from Transformers). I chose these models to evaluate and contrast the effectiveness of traditional machine learning techniques with transformer-based approaches for text classification.

The first model I implemented was the Support Vector Classifier (SVC). SVC is a supervised learning algorithm that finds the optimal approach to separate classes with the maximum margin. For this model, I used the TF-IDF vectors obtained from the preprocessed text as input features. To optimize performance, I performed hyperparameter tuning using a grid search with cross-validation, adjusting parameters such as the regularization strength (C). I chose SVC because it is efficient for high-dimensional data and is known to perform well on text classification tasks, especially when combined with TF-IDF features. During Grid Search for SVC, I used recall as the scoring metric to prioritize capturing as many toxic comments as possible, minimizing false negatives—a critical consideration for content moderation tasks.

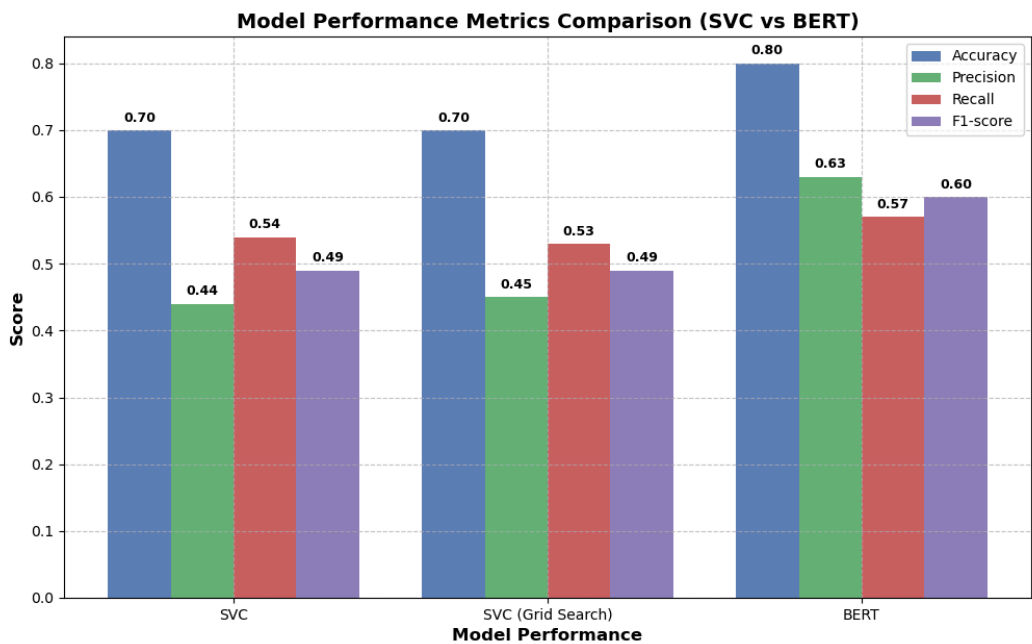
The second model I implemented was BERT, a transformer-based model known for its ability to understand contextual relationships within text. I used a pre-trained BERT model and fine-tuned it on the training dataset. Before training, I tokenized the comments using the BERT tokenizer, which converts text into input IDs and attention masks. During training, I used the AdamW optimizer with cross-entropy loss to minimize classification errors. I also applied techniques such as learning rate scheduling and early stopping to optimize the model's performance and prevent overfitting. I chose BERT because it excels at understanding the nuances of human language, making it particularly effective for detecting toxic comments.

The SVC model provided a fast and interpretable baseline, while the BERT model leveraged deep learning techniques to capture complex language patterns. This comparison allowed me to assess which model was more effective for detecting toxic content in social media comments.

3. Evaluations and Findings

The evaluation of my models was conducted using four essential metrics: accuracy, precision, recall, and F1-score, which are standard for assessing classification performance. Each metric offers a perspective on model effectiveness. Accuracy measures overall correctness of predictions but can be misleading when the classes are imbalanced, as in toxic comment detection. To address this, I focused on precision, which indicates the proportion of predicted toxic comments that were genuinely toxic, helping to reduce false positives. Equally important was recall, which measures how many actual toxic comments were correctly identified, thus minimizing false negatives. Since precision and recall often have a trade-off, I used the F1-score. It represents the harmonic mean of precision and recall, offering a comprehensive measure that balances both aspects of model performance.

I applied these metrics consistently during both training and testing to ensure comprehensive evaluation. During training, I incorporated cross-validation to enhance result reliability and reduce overfitting. Additionally, I performed threshold tuning during testing to adjust the model's sensitivity, exploring different classification thresholds to balance false positives and false negatives. This approach allowed me to select an optimal threshold that aligned with my goal of accurately detecting toxic comments while maintaining a reasonable false-positive rate. Recall was given special attention in this task, as false negatives (toxic comments misclassified as non-toxic) pose a greater risk in real-world moderation systems. However, to maintain a balance between recall and false positives, I used the F1-score as the primary evaluation metric.



The Support Vector Classifier (SVC) was evaluated using accuracy, precision, recall, and F1-score to assess its ability to classify toxic comments effectively. On the training dataset, the SVC model achieved an accuracy of 0.70, precision of 0.44, recall of 0.54, and an F1-score of 0.49, indicating that the model performed well in identifying toxic comments but with some false positives. To improve performance, I applied Grid Search for hyperparameter tuning, which resulted in similar outcomes, with an F1-score of 0.49, suggesting that the initial model was already well-optimized. On the test dataset, I experimented with different classification thresholds to balance precision and recall. The selected threshold of 0.30 produced a balanced outcome, with 165 toxic and 335 non-toxic predictions, aligning closely with the recall achieved during training. Despite extensive hyperparameter tuning, the SVC model's performance remained stable, demonstrating its reliability in detecting toxic content.

The BERT model, a transformer-based deep learning model, was also evaluated using the same metrics. On the test dataset, BERT outperformed the SVC model, achieving an accuracy of 0.80, precision of 0.64, recall of 0.53, and an F1-score of 0.58, highlighting its superior ability

to detect toxic comments while reducing false positives. With 275 toxic and 225 non-toxic predictions, the BERT model demonstrated a stronger balance between precision and recall, benefiting from its deep contextual understanding of language. Despite being more resource-intensive than SVC, BERT's higher accuracy and F1-score illustrate its effectiveness in complex text classification tasks.

During the evaluation of the Support Vector Classifier (SVC), several insights emerged. Firstly, despite applying Grid Search for hyperparameter tuning, the model's performance remained almost same with its initial results, achieving an F1-score of 0.49, indicating that the default hyperparameters were already well-suited for the task. Additionally, adjusting the classification threshold had a significant impact on the balance between precision and recall. Lowering the threshold to 0.20 increased the detection of toxic comments but led to more false positives, while raising it to 0.75 improved precision but severely reduced recall. A threshold of 0.30 provided the most balanced performance, aligning closely with the recall achieved during training. Finally, errors encountered during testing, such as feature dimension mismatches, emphasized the need to apply the same TF-IDF vectorizer to both training and test datasets to maintain consistency. Overall, the SVC model demonstrated stable and reliable performance, with threshold tuning and class balancing playing crucial roles in optimizing results.

The evaluation of the BERT model revealed several key insights that highlighted its strengths and behavior in toxic comment classification. Firstly, BERT outperformed the SVC model across all metrics, achieving a higher accuracy (0.80) and F1-score (0.58), which demonstrates its superior ability to understand complex language patterns and contextual relationships within comments. A notable finding was the model's relatively high precision (0.64) compared to recall (0.53), indicating that BERT was more effective at correctly identifying toxic comments while reducing false positives. Additionally, the model predicted a balanced distribution of 275 toxic and 225 non-toxic comments, closely aligning with the actual class proportions, which suggests strong generalization capabilities.

Interestingly, despite its deep learning architecture, BERT's recall (0.53) was only slightly better than SVC's (0.54), highlighting that while BERT reduces false positives, it may still miss some subtle toxic expressions. This suggests that BERT, although powerful, can benefit from further fine-tuning or additional training data to improve recall. Moreover, the BERT model was more resource-intensive and time-consuming to train compared to SVC, emphasizing the trade-off between performance and computational cost. Overall, the BERT model demonstrated stronger performance in precision and overall accuracy, making it a more reliable choice for detecting harmful content, while also showcasing opportunities for further improvement in recall through additional fine-tuning or ensemble techniques.

Index	Model	Accuracy	Precision	Recall	F1-Score
1	SVC	0.70	0.44	0.53	0.48
2	SVC with GridSearch	0.71	0.45	0.52	0.48
3	BERT	0.80	0.63	0.57	0.60

Through a thorough analysis of evaluation metrics and visual comparisons, BERT stands out as the effective model for detecting toxic comments, providing superior precision and robust generalization. Nevertheless, SVC proves to be a viable choice for a resource-efficient, real-time system, thanks to its performance.

Future Scope:

This report lays a strong foundation for toxic comment detection; however, there are several areas for potential advancement. This project can be expanded in the future by fine-tuning the BERT model with larger and more diverse datasets, thereby improving its ability to detect subtle toxic language. Exploring ensemble methods, such as combining SVC and BERT, could improve performance by taking advantage of their mutual advantages. Furthermore, implementing multilingual models would enable the detection of toxic comments in multiple languages. Lastly, employing alternative interpretability techniques, such as attention visualization or feature importance analysis, could provide deeper insights into the model's decision-making process to make more informed and fair content decisions.

Conclusion:

Summing up, In this project, I developed and compared two machine learning models, SVC and BERT, to detect toxic comments on social media. I started by cleaning the text data and converting it into a format the models could understand. Both models were trained and evaluated using metrics like accuracy, precision, recall, and F1-score. The results showed that BERT performed better, thanks to its ability to understand language context deeply. However, SVC, especially after tuning with Grid Search, proved to be a faster and more efficient option, making it suitable for real-time applications. Through this assignment, I gained valuable experience in building, evaluating, and comparing different models for text classification tasks.

References:

1. Hugging Face. (2023). *BERT Model Documentation*.[\(link\)](#)
2. R. Singh, R. Kashyap and V. Sharma, "Toxic Comment Analyzer using BERT: A Deep Learning Approach for Toxicity Detection," *2023 Second International Conference on Informatics (ICI)*, Noida, India, 2023, pp. 1-6, doi: 10.1109/ICI60088.2023.10421672.[\(link\)](#)
3. Generative Tools: ChatGpt.

Jupyter Notebook link from github:

https://github.iu.edu/tasamud/25SP-ILS-Z639-SMM/blob/main/SocialMediaMining_Assignment1.ipynb